ANALYSIS OF VARIANCE (ANOVA)

ST331

(Statistical Computing 3)

ONE-WAY ANOVA

- One-way ANOVA is used to test the equality of two or more means. Those means are split by a categorical group or factor variable.The explanatory variable may be continuous, discrete, or categorical

$H_0 : \mu_1 = \mu_2 = \ldots = \mu_k$

$H_A : \mu_1, \mu_2, \ldots, \mu_k$ are not all equal

(alternatively, at least one mean differs).

ONE-WAY ANOVA

```
> chick.anova <- aov(weight~feed,data=chickwts)
> summary(chick.anova)
            Df Sum Sq Mean Sq F value   Pr(>F)
feed         5 231129   46226   15.37 5.94e-10 ***
Residuals   65 195556    3009
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

There is **strong statistical evidence** that the type of **feed has a significant effect** on the response variable

ONE-WAY ANOVA

```
> tukey_result <- TukeyHSD(chick.anova)
> tukey_result
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = weight ~ feed, data = chickwts)

$feed
                          diff         lwr       upr     p adj
horsebean-casein    -163.383333 -232.346876 -94.41979 0.0000000
linseed-casein      -104.833333 -170.587491 -39.07918 0.0002100
meatmeal-casein      -46.674242 -113.906207  20.55772 0.3324584
soybean-casein       -77.154762 -140.517054 -13.79247 0.0083653
sunflower-casein       5.333333  -60.420825  71.08749 0.9998902
linseed-horsebean     58.550000  -10.413543 127.51354 0.1413329
meatmeal-horsebean   116.709091   46.335105 187.08308 0.0001062
soybean-horsebean     86.228571   19.541684 152.91546 0.0042167
sunflower-horsebean  168.716667   99.753124 237.68021 0.0000000
meatmeal-linseed      58.159091   -9.072873 125.39106 0.1276965
soybean-linseed       27.678571  -35.683721  91.04086 0.7932853
sunflower-linseed    110.166667   44.412509 175.92082 0.0000884
soybean-meatmeal     -30.480519  -95.375109  34.41407 0.7391356
sunflower-meatmeal    52.007576  -15.224388 119.23954 0.2206962
sunflower-soybean     82.488095   19.125803 145.85039 0.0038845
```

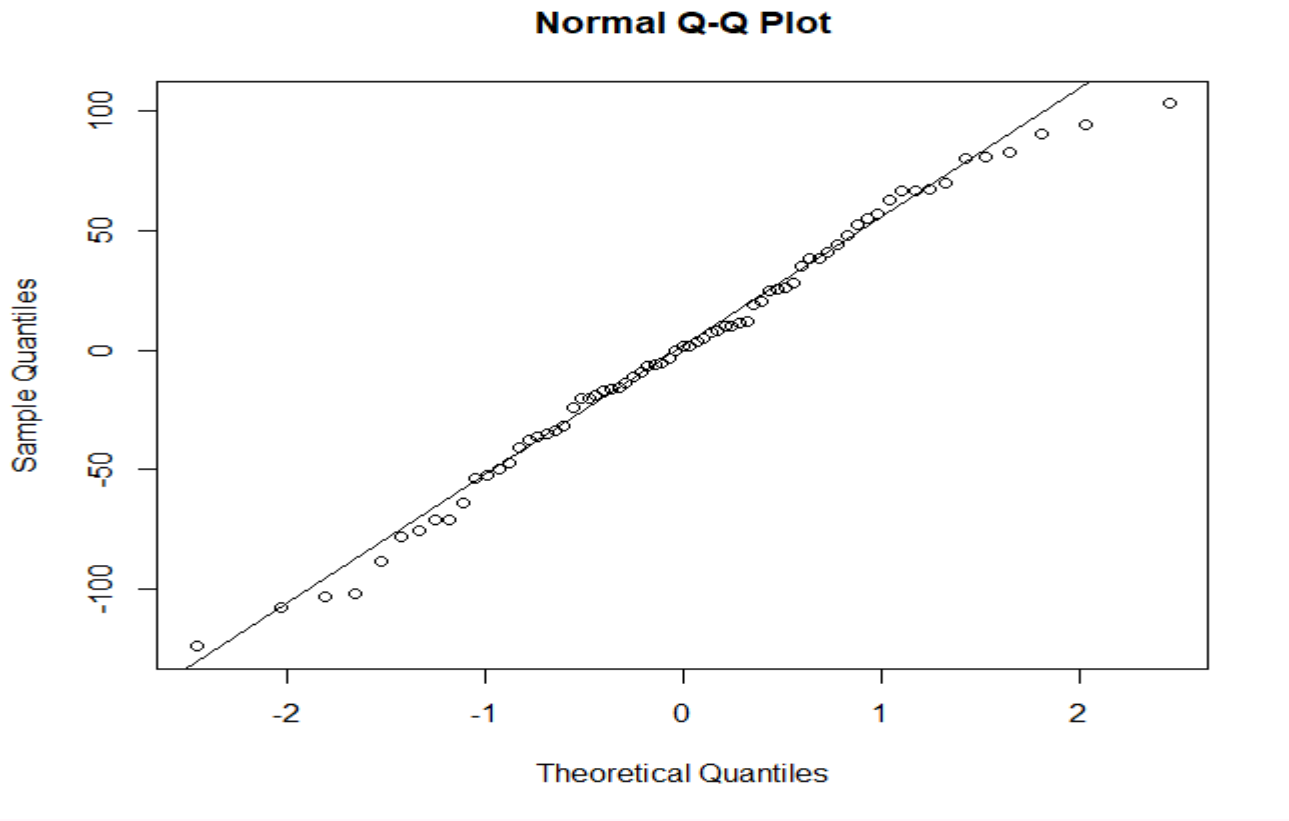
ONE-WAY ANOVA ASSUMPTION

To validate your **ANOVA results**, you need to check the **three major assumptions**:

- **Normality of residuals**
- **Homogeneity of variance**
- **Independence of observations**

NORMALITY OF RESIDUALS

```
> qqnorm(residuals(chick.anova))
> qqline(residuals(chick.anova))
```



NORMALITY OF RESIDUALS

```
> shapiro.test(residuals(chick.anova))

        Shapiro-Wilk normality test

data:  residuals(chick.anova)
W = 0.98616, p-value = 0.6272
```

Decision Rule:

p > 0.05 → residuals are normally distributed

p < 0.05 → normality assumption violated

HOMOGENEITY OF VARIANCE

```
> leveneTest(weight ~ feed, data = chickwts)
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  5  0.7493 0.5896
```

Decision Rule:

$p > 0.05$ → equal variances

$p < 0.05$ → unequal variances

INDEPENDENCE OF ERRORS

```
> durbinWatsonTest(chick.anova)
 lag Autocorrelation D-W Statistic p-value
   1       0.1421075      1.713615   0.058
 Alternative hypothesis: rho != 0
```

Hypotheses

H_0: Residuals are independent ($\rho=0$)

H_1: Residuals are autocorrelated ($\rho\neq0$)

Decision Rule:

$p > 0.05$ → Residuals are independent

$p < 0.05$ → Residuals are autocorrelated

TWO-WAY ANOVA

```
> model<-aov(breaks~wool+tension,data=warpbreaks)
> summary(model)
            Df Sum Sq Mean Sq F value  Pr(>F)
wool         1    451   450.7   3.339 0.07361 .
tension      2   2034  1017.1   7.537 0.00138 **
Residuals   50   6748   135.0
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

TWO-WAY ANOVA WITH INTERACTION

```
> model2<-aov(breaks~wool+tension +wool:tension,data=warpbreaks)
> summary(model2)
            Df Sum Sq Mean Sq F value   Pr(>F)
wool         1    451   450.7   3.765 0.058213 .
tension      2   2034  1017.1   8.498 0.000693 ***
wool:tension 2   1003   501.4   4.189 0.021044 *
Residuals   48   5745   119.7
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

CLASS EXERCISE

Carry out a one-way ANOVA to determine whether the mean **petal width** differs among the three **species** of iris.

If the ANOVA indicates a significant difference, perform a **post-hoc test (Tukey HSD)** to identify which species pairs are responsible for the difference.

Use iris dataset

ST431

(Design and Analysis of Experiment)